POWER BI MINI PROJECT · PROJECT REPORT

Social Media Addiction Dashboard

How students scroll, sleep and study: a six-page interactive report on 705 students in 110 countries.

Project report by **Awanish Verma**

Tool	**Power BI Desktop**
Theme	**Social Pulse**
Report	**6 pages · 82 visuals**
Model	**3 tables · 11 measures**

Overview · Project brief · Dataset · Data model · DAX · Report pages · Interactivity · Findings · Theme · Checklist

Overview

The dashboard analyses how students' social media usage relates to sleep, academic performance, mental health and relationship conflicts. It is built from one Excel workbook and styled with a custom dark theme taken from the colours of the platforms it studies.

Total Students
705
One row per student, no duplicates

Average Usage (hrs/day)
4.92
Range 1.5 to 8.5 hours

% Affected Academically
64.3%
453 of 705 students

Avg Sleep Hours
6.87
Range 3.8 to 9.6 hours

These are the four cards on the Executive Overview page, computed from the full dataset with no filters applied.

Who the report is for

Academic advisors
Usage against academic level and platform. Page 3, Academic Impact.

Mental health counselors
Sleep, mental health and addiction scores. Page 2, Mental Health & Lifestyle.

Institutional researchers
Country and gender breakdowns through the matrix, slicers and story view.

Students & counselors
One student at a glance through the drill-through Student Profile page.

Project brief

The question as set in `Power BI Mini Project.docx`.

Project overview

This project will help students build a full-fledged interactive dashboard in Power BI that analyzes how students' social media usage correlates with sleep patterns, academic performance, mental health, and relationship dynamics.

Stakeholders

- **Academic Advisors** – interested in academic performance and social media usage
- **Mental Health Counselors** – concerned with sleep, mental health, and addiction scores
- **Institutional Researchers** – want country- and gender-level insights
- **Students & Counselors** – view specific profiles for drill-through insights

PHASE 1 Data loading, modeling and cleaning

Task 1: Load the dataset

- Load both sheets from Excel: Student Details and Platform Details
- Check data types and consistency

Task 2: Create relationships

- Join both tables using Student_ID
- Verify a 1:1 relationship
- Ensure both tables are in Star Schema (fact = behavior metrics, dimension = demographics)

Task 3: Add a custom date table (for mock date-based analysis)

- Create a simple date table using CALENDAR() for future use
- Mark as a date table

```
DateTable = CALENDAR(DATE(2023, 1, 1), DATE(2023, 12, 31))
```

PHASE 2 Data preparation and transformation

Task 4: Create calculated columns

```
Health Band =
IF(Mental_Health_Score >= 7, "Good",
   IF(Mental_Health_Score >= 4, "Average", "Poor"))

Conflict Level =
SWITCH(TRUE(),
   Conflicts_Over_Social_Media > 5, "High",
   Conflicts_Over_Social_Media > 2, "Medium",
   "Low")
```

Task 5: Create measures

```
Avg Sleep Hours =
AVERAGE(Student_Details[Sleep_Hours_Per_Night])

Avg Usage Hours =
AVERAGE(Platform_Details[Avg_Daily_Usage_Hours])

% Affected Academically =
DIVIDE(
   CALCULATE(COUNTROWS(Platform_Details),
      Platform_Details[Affects_Academic_Performance]
         = "Yes"),
   COUNTROWS(Platform_Details))

Addicted Student Count =
CALCULATE(COUNTROWS(Student_Details),
   Student_Details[Addicted_Score] > 7)
```

PHASE 3 Time intelligence (simulation)

Task 6: Use date table (optional time-based extension)

```
Monthly Trend =
CALCULATE([Avg Usage Hours], DATESYTD(DateTable[Date]))
```

PHASE 5 Interactivity and storytelling

Task 7: Slicers & filters

Add slicers for:

- Mental Health & Lifestyle – Gender
- Relationships & Conflicts – Country, Academic Level
- Academic Impact – Age Range (using numeric slider)

PHASE 4

Report pages and visuals

Page 1: Executive Overview

- Card visuals: Total Students, Average Usage, % Affected Academically, Avg Sleep Hours
- Bar chart: Addiction Score by Academic Level
- Line chart: Avg Usage Hours by Age
- Pie chart: Gender Distribution

Page 2: Mental Health & Lifestyle

- Matrix: Country vs Health Band
- Scatter chart: Addicted Score vs Mental Health Score
- Line chart: Sleep Hours vs Age

Page 3: Academic Impact

- Bar chart: Avg Daily Usage vs Academic Level
- Column chart: Academic Performance Impact by Platform

Page 4: Relationships and Conflicts

- Stacked bar: Conflict Level by Relationship Status
- Table: Student wise Addiction Score
- Pie: Relationship Status Distribution

Page 5: Interactive Story View

- Bookmark View 1, Social Media by Gender. Bar chart: Avg_Daily_Usage_Hours by Gender. Donut chart: Most_Used_Platform by Gender
- Bookmark View 2, Social Media by Academic Level. Bar chart: Avg_Daily_Usage_Hours by Academic_Level. Pie chart: Most_Used_Platform by Academic_Level
- Go to View > Bookmarks Pane and Selection Pane
- For Bookmark 1 (Gender View): hide the Academic Level visuals, show the Gender visuals, click "Add", name it Gender View
- For Bookmark 2 (Academic Level View): do the opposite, then click "Add" and name it Academic Level View
- Create two buttons to show Gender View and Academic Level View

Page 6: Drill-through page, Student Profile

- Create a new page for drill-through to Student_ID
- Drill-through on Student_ID
- Show: Student Demographics, Sleep Hours, Usage Hours, Addicted Score, Most Used Platform, Conflicts, Mental Health Rating

Final deliverables

Students should submit the Power BI .pbix file compressed in a zip folder. The dashboard submitted should be fully functional.

Dataset

`Dataset/Students Social Media Addiction.xlsx` holds two sheets with 705 rows each, joined on `Student_ID`. Neither sheet has blank cells or duplicate IDs, and every student appears in both.

Student Details 10 COLUMNS

Column	Type	Values in the data
`Student_ID`	Whole number	Key, 705 unique
`Age`	Whole number	18 to 24
`Gender`	Text	Female 353, Male 352
`Academic_Level`	Text	Undergraduate 353, Graduate 325, High School 27
`Country`	Text	110 countries; India 53, USA 40, Canada 34
`Sleep_Hours_Per_Night`	Decimal	3.8 to 9.6, mean 6.87
`Mental_Health_Score`	Whole number	4 to 9, mean 6.23
`Relationship_Status`	Text	Single 384, In Relationship 289, Complicated 32
`Conflicts_Over_Social_Media`	Whole number	0 to 5, mean 2.85
`Addicted_Score`	Whole number	2 to 9, mean 6.44

Platform Details 4 COLUMNS

Column	Type	Values in the data
`Student_ID`	Whole number	Key, 705 unique
`Avg_Daily_Usage_Hours`	Decimal	1.5 to 8.5, mean 4.92
`Most_Used_Platform`	Text	12 platforms; Instagram 249, TikTok 154, Facebook 123
`Affects_Academic_Performance`	Text	Yes 453, No 252

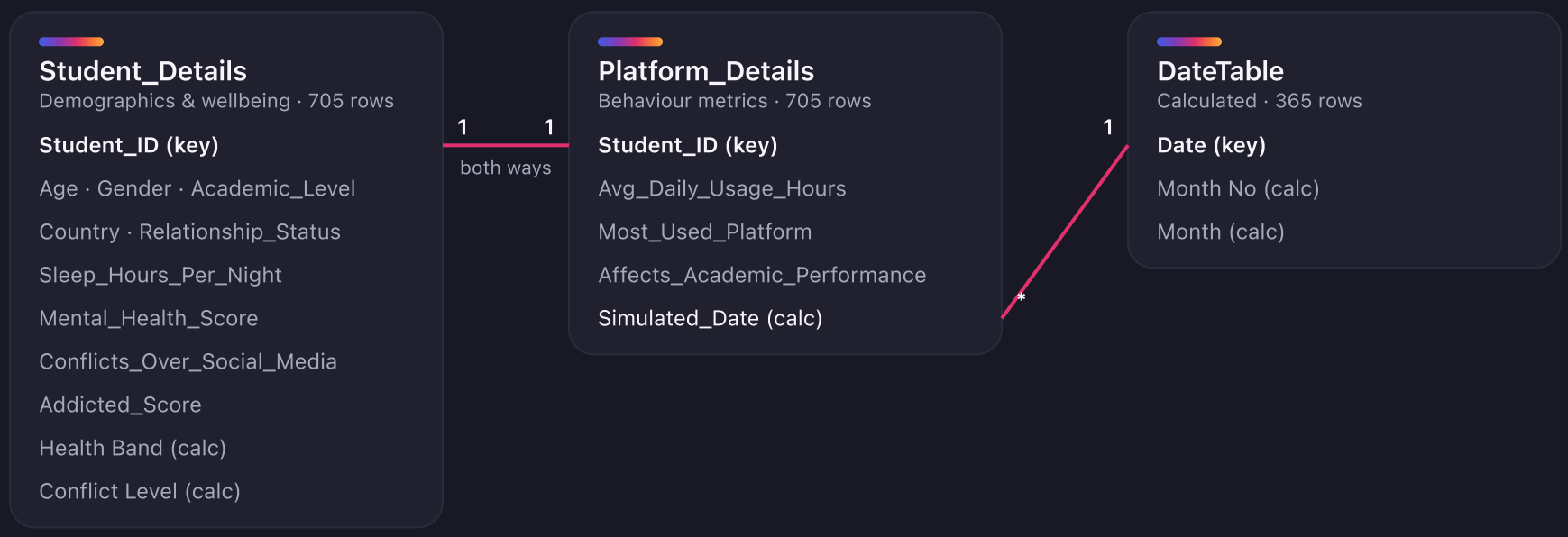
Cleaning steps in Power Query

- Promote the first row to headers.
- Set an explicit data type on every column.
- Trim whitespace from all text columns.
- Remove rows with a blank `Student_ID`.
- Remove duplicate `Student_ID` rows.

The last two steps change nothing on the current file. They keep the 1:1 relationship valid if the workbook is replaced.

Data model

Three tables. `Platform_Details` carries the behaviour metrics and `Student_Details` the demographics and wellbeing scores. A calculated `DateTable` supports the simulated time analysis.



Student ↔ Platform

One to one on `Student_ID`, cross-filtering in both directions, so a slicer on gender filters platform measures and a platform selection filters student counts.

Platform → Date

Many to one from `Simulated_Date` to `DateTable[Date]`. The table is flagged as the model's time table and has a `Month` column sorted by `Month No`.

DatasetPath parameter

Both queries read the workbook from one text parameter, so the report can be pointed at the Excel file on any machine without editing a query.

DAX

Two calculated columns band the raw scores, one mocks a date, and eleven measures feed the visuals.

Calculated columns and table

Health Band — Student_Details

```
Health Band =
IF(Student_Details[Mental_Health_Score] >= 7,
   "Good",
   IF(Student_Details[Mental_Health_Score] >= 4,
      "Average", "Poor"))
```

Groups the 1–10 mental health score into three bands for the matrix.

Conflict Level — Student_Details

```
Conflict Level =
SWITCH(TRUE(),
   Student_Details[Conflicts_Over_Social_Media] > 5,
      "High",
   Student_Details[Conflicts_Over_Social_Media] > 2,
      "Medium",
   "Low")
```

Bands the conflict count for the stacked bar on page 4.

DateTable — Calculated table

```
DateTable = CALENDAR(DATE(2023, 1, 1), DATE(2023, 12,
31))
Month No  = MONTH(DateTable[Date])
Month     = FORMAT(DateTable[Date], "MMM")
```

365 days of 2023, with month columns for the trend chart axis.

Simulated_Date — Platform_Details

```
Simulated_Date =
DATE(2023, 1, 1)
   + MOD(Platform_Details[Student_ID] * 37, 365)
```

The dataset has no dates. This spreads students across 2023 so the time-intelligence measure has something to plot. It is not real data.

Measures required by the brief

Avg Sleep Hours — 0.00

```
AVERAGE(Student_Details[Sleep_Hours_Per_Night])
```

Avg Usage Hours — 0.00

```
AVERAGE(Platform_Details[Avg_Daily_Usage_Hours])
```

% Affected Academically — 0.0%

```
DIVIDE(
   CALCULATE(COUNTROWS(Platform_Details),
      Platform_Details[Affects_Academic_Performance]
         = "Yes"),
   COUNTROWS(Platform_Details))
```

Addicted Student Count — #,0

```
CALCULATE(COUNTROWS(Student_Details),
   Student_Details[Addicted_Score] > 7)
```

199 students score 8 or 9.

Monthly Trend — 0.00

```
CALCULATE([Avg Usage Hours], DATESYTD(DateTable[Date]))
```

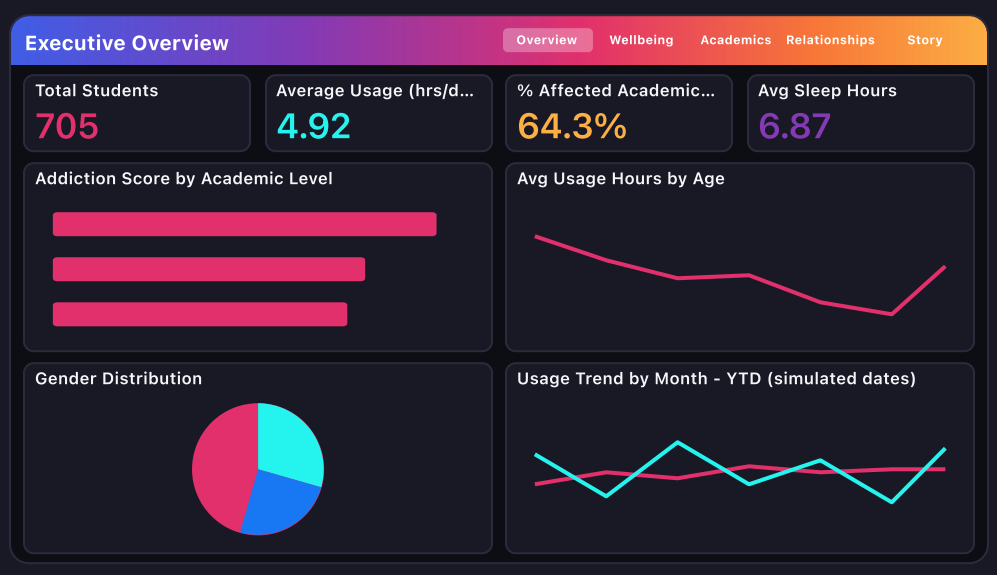
Year-to-date average usage over the simulated dates.

Supporting measures

Measure	Table	Expression	Used on
Total Students	Student_Details	`COUNTROWS(Student_Details)`	Cards, pies, matrix, column chart
Avg Addicted Score	Student_Details	`AVERAGE(Student_Details[Addicted_Score])`	Bar chart, scatter, profile
Avg Mental Health Score	Student_Details	`AVERAGE(Student_Details[Mental_Health_Score])`	Card, scatter
Avg Conflicts	Student_Details	`AVERAGE(Student_Details[Conflicts_Over_Social_Media])`	Profile card
Mental Health Rating	Student_Details	Health band and score as text, e.g. `Good (8/10)`, when one student is selected	Profile card
Top Platform	Platform_Details	`SELECTEDVALUE(Platform_Details[Most_Used_Platform], "Multiple")`	Profile card

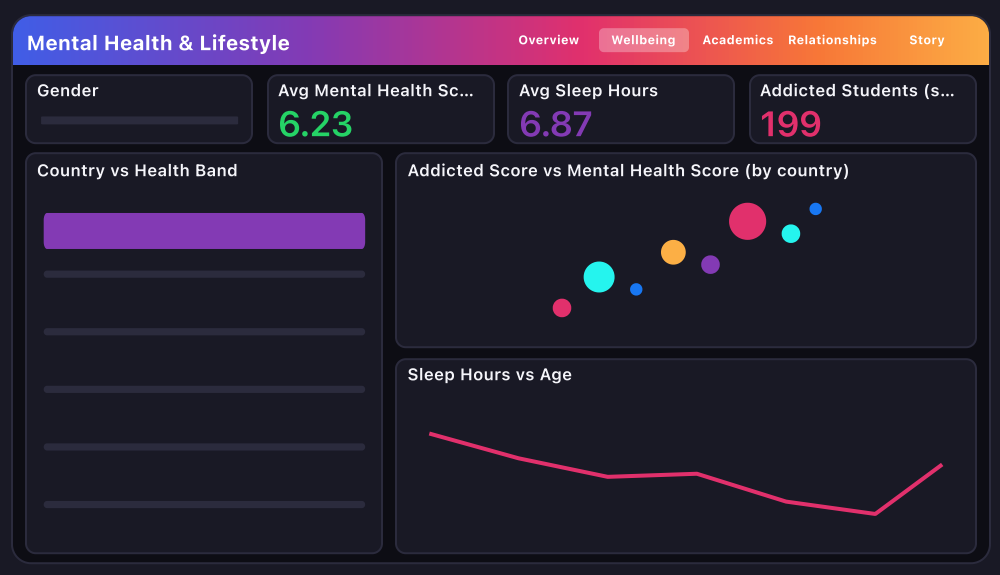
Report pages

Every page is 1280 × 720 with the same gradient header and five navigation buttons. The layouts below are drawn from the report's actual visual positions.



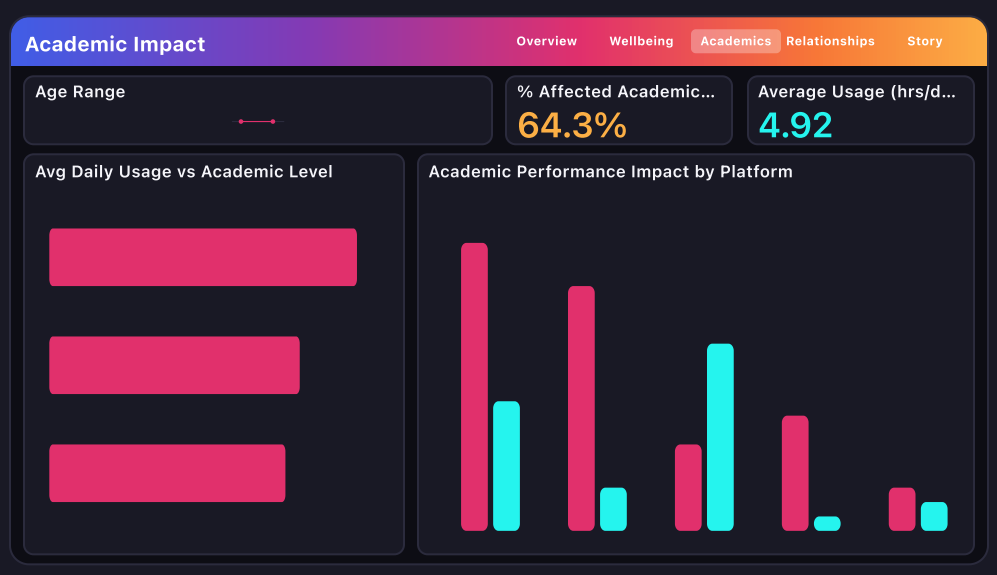
1. Executive Overview How students scroll, sleep and study

Four headline cards, then addiction by academic level, usage by age, the gender split and the simulated monthly trend.



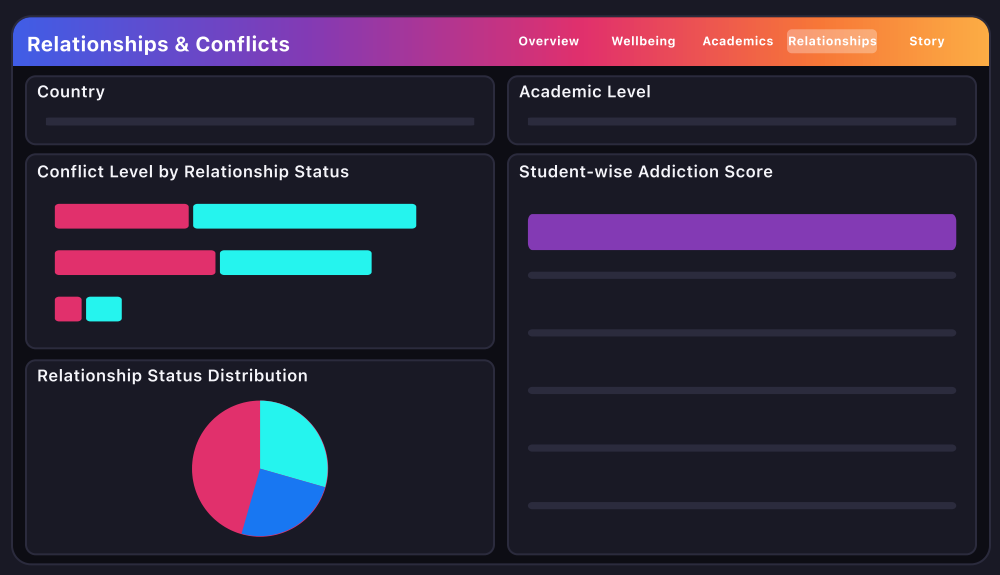
2. Mental Health & Lifestyle Sleep, wellbeing and addiction signals

A gender slicer and three cards over a country-by-health-band matrix, a scatter of addiction against mental health, and sleep by age.



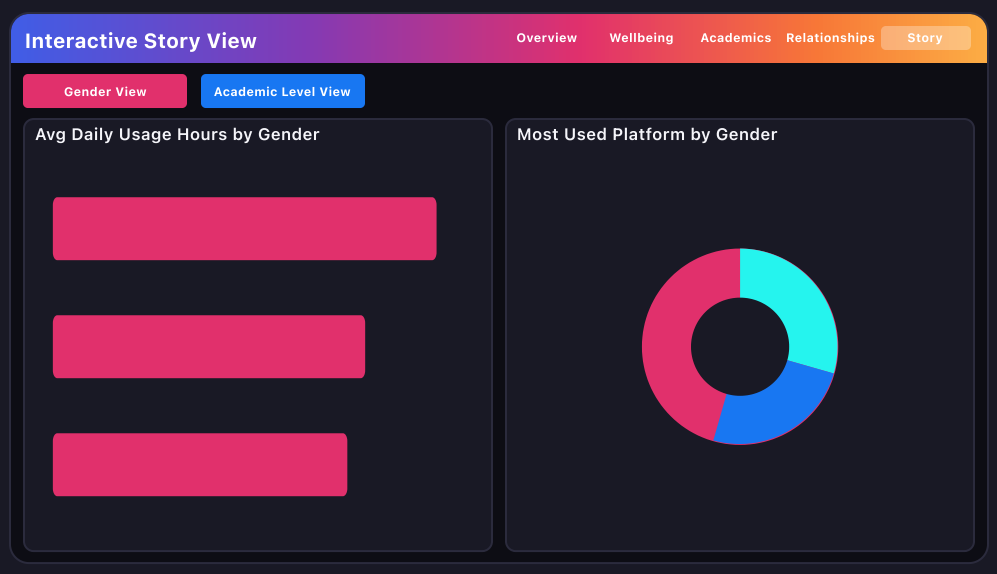
3. Academic Impact Does screen time cost grades?

An age range slider and two cards over usage by academic level and the academic impact of each platform.



4. Relationships & Conflicts When the feed starts the fight

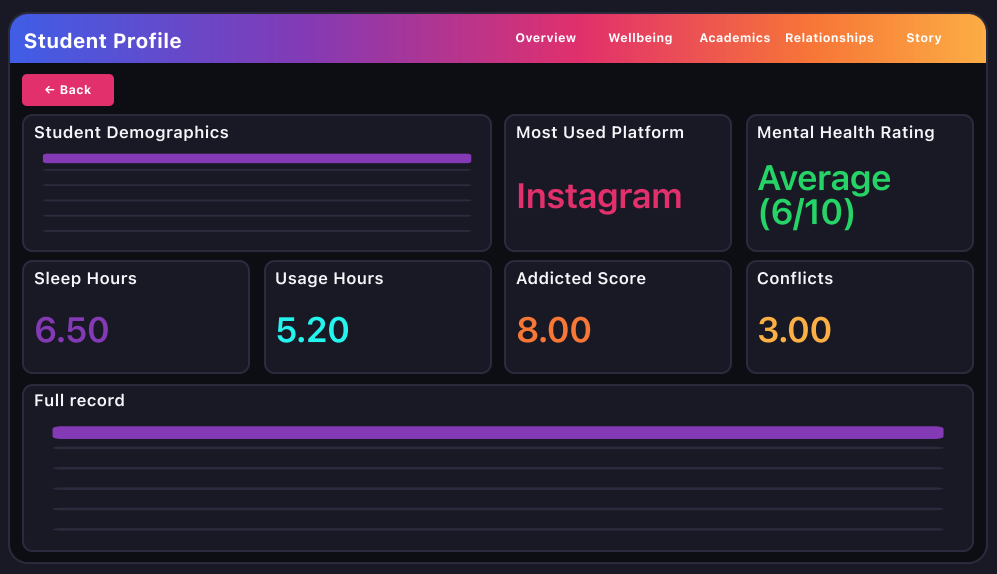
Country and academic level slicers over conflict level by relationship status, the relationship status split, and the student table that drills through to a profile.



5. Interactive Story View

Tap a story to switch the lens

Two bookmark buttons swap the charts between a gender view (shown) and an academic level view, which sits hidden in the same two slots.



6. Student Profile

Drill-through: one student at a glance

Reached by drill-through on Student_ID. Demographics, six cards for the selected student, the full record and a Back button. Values shown are for student 1.

Interactivity

Slicers

Page	Field	Style
Mental Health & Lifestyle	Gender	Dropdown
Academic Impact	Age	Numeric range slider
Relationships & Conflicts	Country	Dropdown
Relationships & Conflicts	Academic Level	Dropdown

Bookmarks

The Story page stacks two pairs of charts in the same two slots. Each bookmark shows one pair and hides the other, without changing any filters.

- **Gender View**: usage by gender (bar) and platform by gender (donut).
- **Academic Level View**: usage by academic level (bar) and platform by academic level (pie).

Two buttons above the charts switch between them.

Drill-through

Student Profile is a drill-through page on `Student_ID`. On page 4, right-click a row in the Student-wise Addiction Score table, choose Drill through, then Student Profile. The Back button returns to the previous page.

Page navigation

The header buttons (Overview, Wellbeing, Academics, Relationships, Story) are page-navigation actions. The button for the current page is highlighted. In Power BI Desktop, hold Ctrl while clicking a button.

Findings

What the dashboard shows on the full dataset. All figures are computed from the workbook.

High school students score highest on addiction

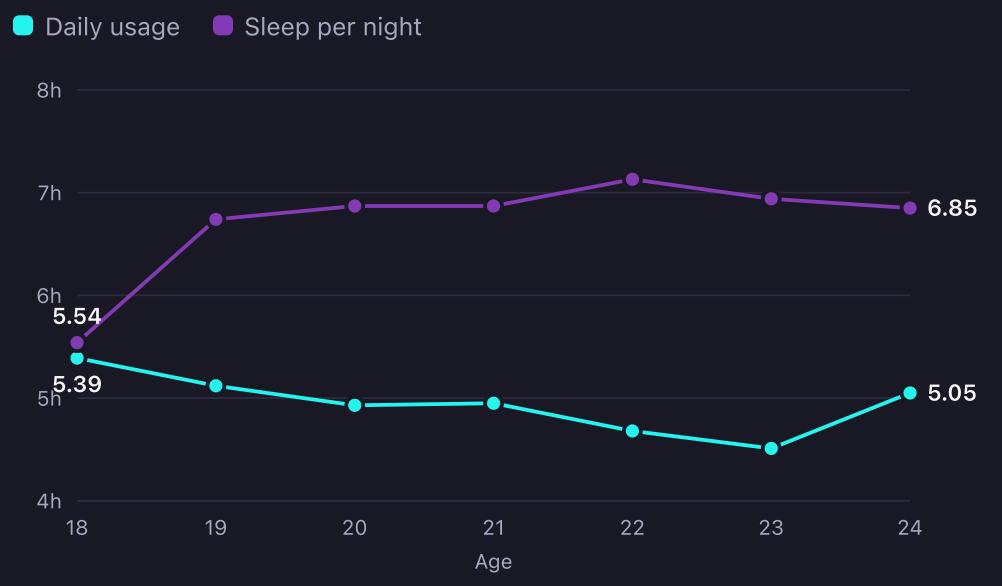
Average addicted score (2–9 scale) by academic level



High school is a small group (27 students), so treat its average with care.

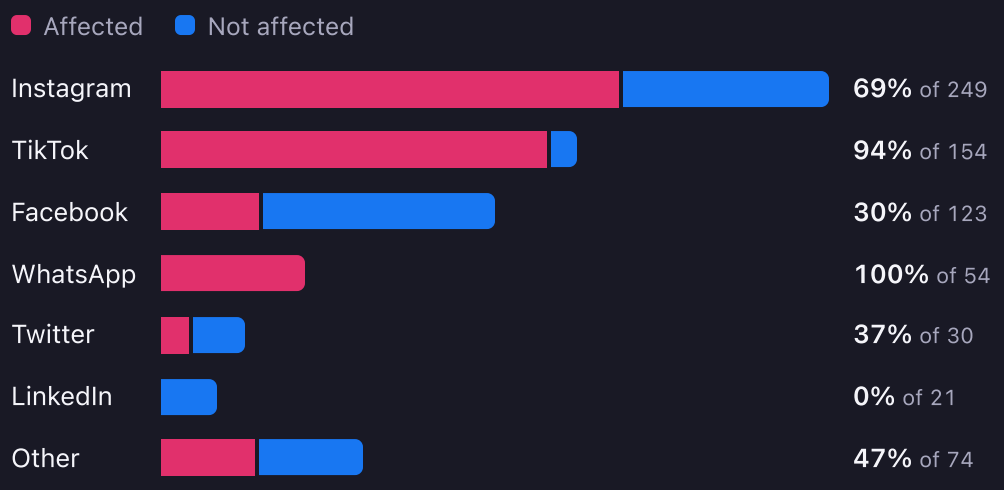
The youngest students use more and sleep less

Average hours per day by age



TikTok and WhatsApp users report the most academic impact

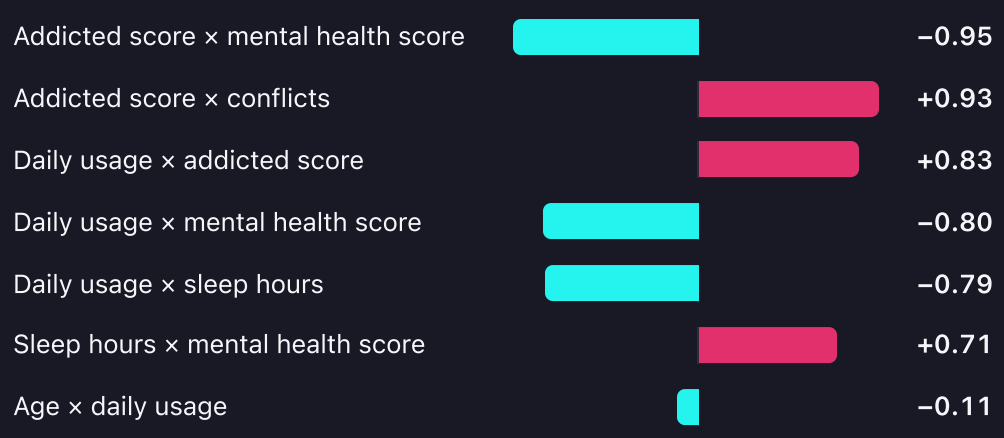
Students by most-used platform, split by whether usage affects their academic performance



"Other" combines KakaoTalk, LINE, Snapchat, VKontakte, WeChat and YouTube (74 students).

The scores move together

Correlation between pairs of measures, from −1 to +1



Correlations this strong are unusual in survey data. They describe this dataset and do not show cause.

5.54 vs 3.80 hours

Students who say social media affects their studies average 5.54 hours a day. Those who say it does not average 3.80.

6.34 vs 7.81 hours of sleep

The affected group sleeps about an hour and a half less per night than the unaffected group.

199 students above 7

28% of students have an addicted score of 8 or 9, the dashboard's definition of an addicted student.

Things to know when reading the report

- The **Poor** health band never appears. The lowest mental health score in the data is 4, so every student is Average (421) or Good (284).
- The **High** conflict level never appears. The highest conflict count is 5, so every student is Low (255) or Medium (450).
- The monthly trend chart uses **simulated dates** derived from `Student_ID`. Its shape carries no real meaning; it is there to demonstrate `DATESYTD`.
- The scatter chart plots one bubble per **country** (average addicted score against average mental health score, sized by student count), not one per student.
- With 110 countries, most cells in the Country vs Health Band matrix hold only a few students.

Social Pulse theme

The report uses a custom theme file, `Social Pulse Theme.json`. Its data colours are borrowed from social platform brand colours, on dark panels with 14px rounded borders. The header gradient is a 2560 × 128 image generated by the build script.

Pink #E1306C Data 1 · accent	**Cyan** #25F4EE Data 2	**Blue** #1877F2 Data 3	**Yellow** #FCAF45 Data 4 · neutral	**Purple** #833AB4 Data 5 · table header	**Green** #25D366 Data 6 · good	**Orange** #F77737 Data 7	**Page** #0D0D14 Background
Panel #191925 Visual background	**Border** #2B2B3D Borders, grid	**Text** #F5F5FA Titles, values	**Muted** #A7A7BD Labels				

Header gradient: `#405DE6` → `#833AB4` → `#E1306C` → `#F77737` → `#FCAF45`

Type is Segoe UI for labels and Segoe UI Semibold for titles (12pt) and card callouts. Tables and the matrix use a purple header row with alternating panel shades. Line charts use 3px strokes with markers.

Requirements checklist

Each task from the brief and where it is implemented.

Phase	Requirement	Implementation	Status
1	Load both Excel sheets, check types	Two Power Query queries with explicit types and trimmed text	DONE
1	Relate tables on Student_ID, 1:1	One-to-one relationship, both directions	DONE
1	Custom date table with CALENDAR()	`DateTable`, 1 Jan – 31 Dec 2023, flagged as the time table	DONE
2	Health Band, Conflict Level columns	Calculated columns on `Student_Details`	DONE
2	Four measures	Avg Sleep Hours, Avg Usage Hours, % Affected Academically, Addicted Student Count	DONE
3	Monthly Trend with DATESYTD	Measure plus a trend line chart on page 1, over simulated dates	DONE
4	Page 1: 4 cards, bar, line, pie	All present, plus the monthly trend chart	DONE
4	Page 2: matrix, scatter, line	All present. Scatter is aggregated by country	NOTE
4	Page 3: bar, column	Usage by academic level; platform by academic impact	DONE
4	Page 4: stacked bar, table, pie	All present	DONE
4	Page 5: two bookmark views, two buttons	Gender View and Academic Level View	DONE
4	Page 6: drill-through on Student_ID	Student Profile with demographics, sleep, usage, addicted score, platform, conflicts, mental health rating	DONE
5	Slicers: gender, country, academic level, age range	On pages 2, 4 and 3 as specified	DONE
—	Submit a .pbix in a zip	Open the .pbit, load the data, save as .pbix, then zip	TO DO

Project report by Awanish Verma. Figures on this page were computed directly from the Excel workbook. The report itself is generated by `build_powerbi.py`; page layouts and DAX shown here are taken from that script and its output.